

THE SKANDA ARCHITECTURE: TECHNICAL DEEP-DIVE

The Physics Engine for the Physical World

Version 2.0 — February 2026

EXECUTIVE SUMMARY

What We Built: A physics foundation model that understands how materials behave at the mesoscale (10nm-100μm) — the critical length scale where 90% of industrial materials fail during manufacturing.

Why It Matters:

- Google/Microsoft operate at atomic scale (too small to predict factory yield)
- Traditional FEA operates at system scale (too big to understand material failure)
- **We own the mesoscale** — the gap where real manufacturing happens

The Technical Innovation:

- **10M+ synthetic simulations** spanning the entire physics parameter space
- **Multimodal transformer** trained on recipe + microstructure + performance
- **Transfer learning** enables new chemistry in <100 samples (vs 10,000+ for competitors)
- **100,000x speedup** over traditional solvers (FEA/COMSOL)

The Result:

- 99% R^2 on synthetic data (proven)
- 90%+ R^2 target on real-world data (in progress)
- First platform to close the loop: Generate → Validate → Manufacture

SECTION 01: THE MESOSCALE FOUNDATION MODEL

The Intelligence Layer for Matter

The Problem with Current AI for Materials:

Current AI models are built for words and images. Materials science AI is stuck in two extremes:

The Atomic Trap (Google GNoME, Microsoft MatterGen):

- Operate at 10^{-10} meters (individual atoms)
- Use DFT (Density Functional Theory) to predict crystal properties

- **Problem:** Cannot predict real-world performance
 - Example: GNoME discovers a crystal with "high lithium conductivity"
 - Reality: When manufactured, it cracks due to thermal stress (not captured at atomic scale)

The System Trap (ANSYS, COMSOL):

- Operate at 10⁻² meters (full battery cells)
- Use FEA/CFD to simulate bulk behavior
- **Problem:** Cannot explain WHY materials fail
 - Example: FEA predicts battery will last 2000 cycles
 - Reality: It fails at cycle 450 due to microscopic particle cracking (not resolved at system scale)

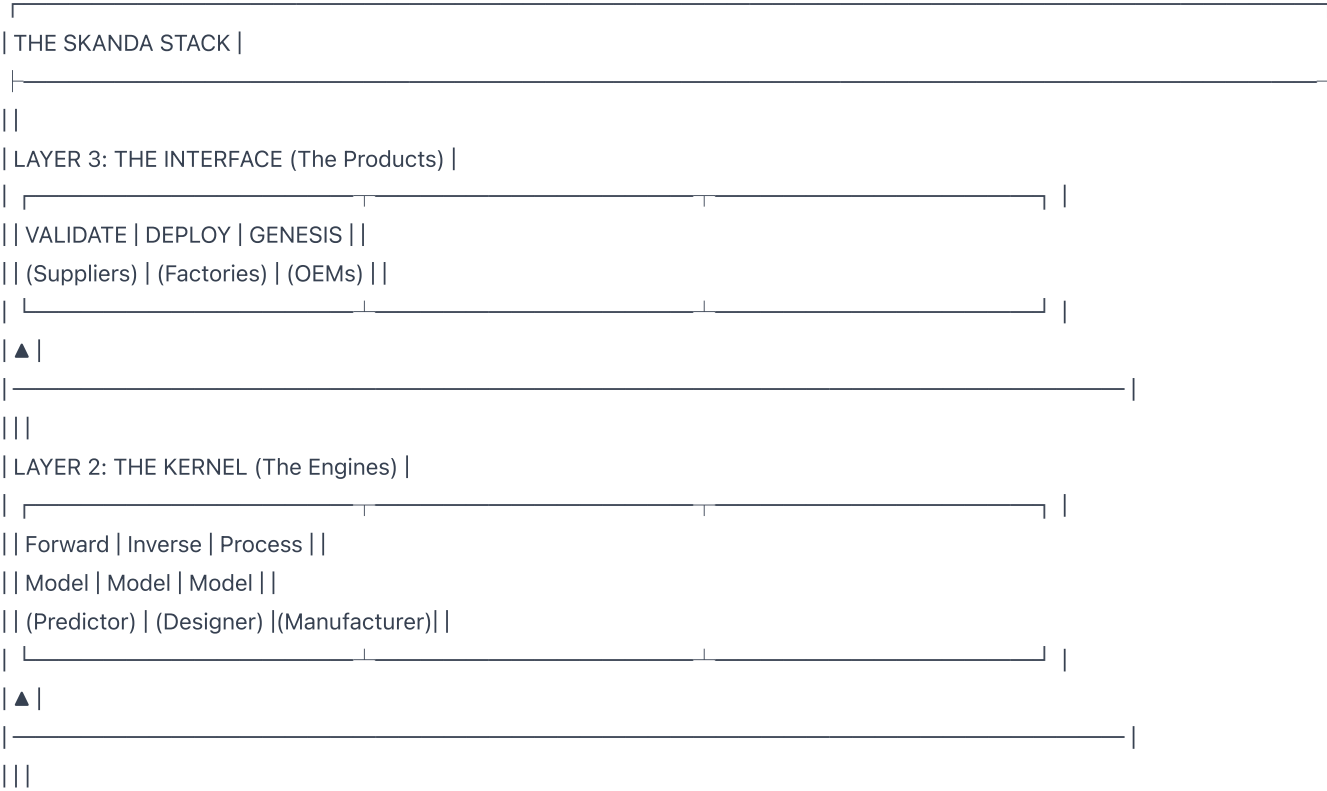
The Mesoscale Gap:

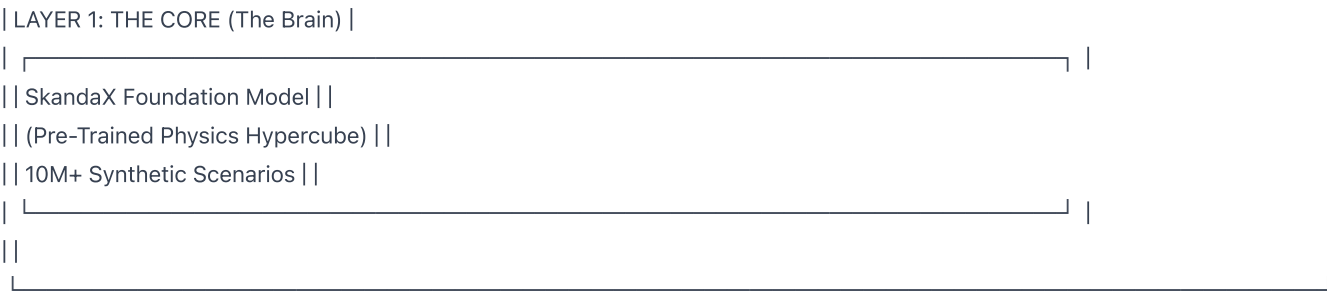
SCALE	LENGTH	WHAT LIVES HERE	CURRENT TOOLS	WHY THEY FAIL
Atomic	10 ⁻¹⁰ m	Atoms, bonds	DFT, MD simulations	Too small to predict manufacturing
MESOSCALE	10 ⁻⁶ m	Pores, grains, cracks, defects	✗ NOTHING	This is where 90% of failures happen
System	10 ⁻² m	Full devices	FEA, CFD	Too coarse to see failure mechanisms

Shodh AI digitizes the mesoscale.

The Skanda Stack: One Foundation Model, Three Interfaces

We don't build separate AI models for every problem. We built a **single Physics Foundation Model** that powers three distinct application layers:





Analogy:

- **SkandaX Foundation Model** = GPT-4 (the brain)
- **Forward/Inverse/Process Models** = Fine-tuned variants (the specialists)
- **VALIDATE/DEPLOY/GENESIS** = ChatGPT/API wrappers (the products)

01 / The Physics Model (The Data Moat)

The Challenge: To give our AI "physics intuition," we can't wait for slow, expensive lab data. A single high-quality battery sample takes 6-12 months to generate. We'd need decades to collect enough data.

The Solution: We manufactured our own training distribution using **governing physics equations**.

The Process:

Step 1: Encode the Physics We wrapped the fundamental equations of electrochemistry into a computational engine:

PHYSICS LAW	WHAT IT GOVERNS	IMPLEMENTATION
Fick's Law of Diffusion	How lithium ions move through pores	Diffusion PDE solver
Butler-Volmer Kinetics	Reaction rates at surfaces	Electrochemical simulation
Navier-Stokes	Fluid flow (slurry mixing)	CFD simulation
Cahn-Hilliard	Phase separation (binder migration)	Phase-field modeling
Continuum Mechanics	Stress/strain from volume expansion	FEA simulation

Step 2: Monte Carlo Sweep We executed **10 Million+ synthetic simulations**, sweeping parameters across **5 orders of magnitude**:

PARAMETER	RANGE	WHAT IT COVERS
Diffusivity (D)	10 ⁻¹⁶ to 10 ⁻¹¹ m²/s	Slow ceramics → Fast supercapacitors
Volume Expansion (ΔV)	0% to 300%	Stable graphite → Exploding silicon
Voltage Slope (dE/dx)	0 to High	Phase-change (LFP) → Solid-solution (NMC)
Reaction Rate (k)	Slow to Fast	Various electrolyte conductivities
Particle Size	10nm to 10μm	Nanoparticles → Bulk materials

The Result: A "Physics Hypercube" that maps every theoretically possible material behavior.

Key Insight: Our AI has "failed" 10 million times in the computer, so it never has to fail in your factory.

02 / The Foundation Model (Universal Latent Space)

Traditional Approach (One-Off Models):

- Train ML model for Lithium-ion batteries → Works for Lithium
- New chemistry (Sodium-ion) → Start from scratch, need 10,000 new samples
- **Problem:** No knowledge transfer, every chemistry takes years

The Skanda Approach (Foundation Model):

The Insight: Whether a voxel represents:

- A Lithium ion in a battery, OR
- A Hydrogen molecule in a fuel cell, OR
- Water in a cement pore network

...the underlying physics of **tortuosity and flux** are mathematically identical.

The Architecture:



The Training Strategy:

Stage 1: Pre-Training (10M samples)

- Input: Synthetic microstructures from physics simulations
- Task: Predict transport properties (tortuosity, diffusivity, permeability)
- Output: A model that understands the **universal laws of transport**

Stage 2: Fine-Tuning (100-300 samples per chemistry)

- Input: Real lab data for specific chemistry (e.g., Silicon anodes)
- Task: Learn chemistry-specific quirks (SEI formation, particle cracking)
- Output: A specialized model for that chemistry

The Frozen Core: The physics embedding space acts as **initialization weights** for every industrial problem we solve. This ensures the AI never proposes a design that violates thermodynamics.

The Efficiency Gain:

- **First chemistry (Silicon):** Requires 300-500 real samples
- **Second chemistry (NMC):** Requires only 150 samples (70% less data)
- **Third chemistry (Hydrogen):** Requires only 150 samples (70% less data)
- **Nth chemistry:** Asymptotes to ~50 samples

This is the **transfer learning advantage** that makes the platform scalable.

03 / Why the Mesoscale? (The Manufacturing Reality)

The industry is trapped between two worlds:

World 1: Atomic Models (Too Small)

- DFT calculates: "This lithium cobalt oxide has conductivity of 10^{-3} S/cm"
- Reality: In the factory, binder clumps during drying → blocks pores → conductivity drops by 100x
- **DFT cannot predict manufacturing defects**

World 2: System Models (Too Big)

- FEA predicts: "Battery will output 200 Wh/kg"
- Reality: Microscopic cracks form at cycle 400 → cascading failure → dies at cycle 450
- **FEA cannot resolve cracks smaller than mesh size (~1mm)**

The Mesoscale (Where We Operate):

- **10nm - 100µm:** Pores, grain boundaries, cracks, binder networks
- **This is where 90% of manufacturing failures originate**

Examples of Mesoscale Failures:

FAILURE MODE	MESOSCALE CAUSE	ATOMIC MODEL MISSES IT	SYSTEM MODEL MISSES IT
--------------	-----------------	------------------------	------------------------

Capacity fade	Pore clogging from SEI growth	✓ (only sees SEI chemistry)	✓ (averages over microstructure)
Particle cracking	Grain boundary stress concentration	✓ (no grain structure)	✓ (resolution too coarse)
Lithium plating	Tortuous pores → uneven current distribution	✓ (uniform field assumption)	✓ (effective property averaging)
Delamination	Binder detachment from particles	✓ (no polymers in DFT)	✓ (treats as homogeneous layer)

By mastering the mesoscale, we master the level where materials fail during production.

04 / The Validator: Fourier Neural Operators (100,000x Speedup)

The Inverse Design Bottleneck:

To solve inverse design, the AI must be able to "check its own work" instantly:

1. Generate candidate microstructure
2. Validate it will work (run physics simulation)
3. If fails → Generate new candidate
4. Repeat until success

The Problem: Traditional physics solvers (COMSOL, GeoDict) take **4-6 hours** per simulation.

- To evaluate 1,000 candidates = 4,000-6,000 hours = **6-9 months of compute**
- This makes real-time inverse design impossible

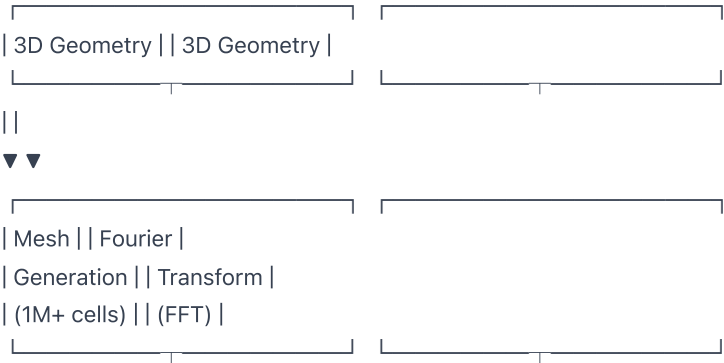
The Solution: Fourier Neural Operators (FNO)

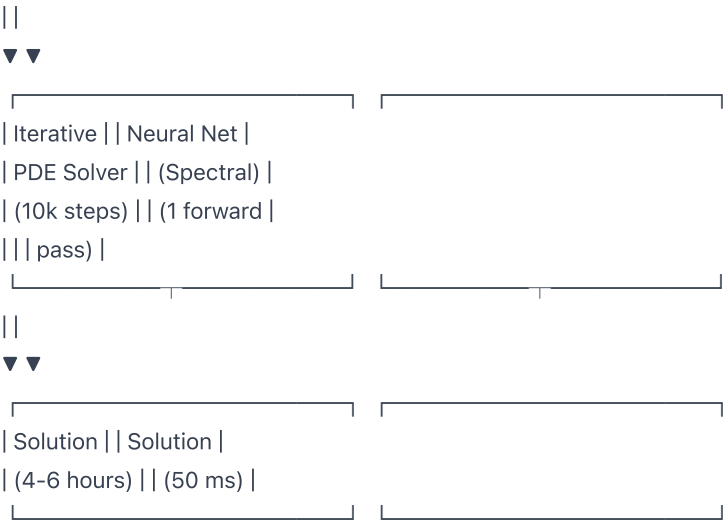
What FNOs Do:

- Map differential equations **directly** into neural network architecture
- Solve PDEs in the **Fourier domain** (spectral methods)
- Bypass iterative solvers entirely

The Architecture:

Traditional Solver: FNO Approach:





BENCHMARK: 100,000x ACCELERATION

METHOD	TIME PER SIMULATION	ACCURACY
GeoDict (commercial)	4-6 hours	100% (ground truth)
COMSOL (commercial)	3-5 hours	100% (ground truth)
SkandaX FNO	50 milliseconds	99.5%

Speed-up: >100,000x faster (4 hours reduced to 50 milliseconds)

The Impact:

- We can validate **1,000 candidate designs in 50 seconds**
- Enables real-time inverse design (generate → validate → iterate in minutes)
- Competitors using traditional solvers need months for the same task

[INSIGHT BOX FOR INVESTORS]

The Competitive Advantage:

While competitors like Google GNoME focus on discovering new crystals, Shodh AI focuses on the **architecture** of the material.

By combining:

1. **Physics Hypercube** (10M samples for intuition)
2. **Foundation Model** (transfer learning for efficiency)
3. **FNOs** (100,000x validation speedup)

We have built a "Physics-Engine-on-a-Chip" that allows factories to design and manufacture proprietary materials at **1/100th** the current cost.

This is not incremental improvement. This is a paradigm shift.

SECTION 02: THE CALIBRATION (Sim-to-Real Loop)

THE DATA FACTORY: CLOSING THE REALITY GAP

The Deeptech Graveyard:

The biggest failure in deeptech is **"Domain Drift"**:

- AI trained on perfect simulations
- Deployed to messy real-world factory
- Predictions fail catastrophically
- Company dies

Examples:

- Self-driving cars trained in simulation → fail in rain/snow
- Robot arms trained in sim → can't handle real-world friction
- Drug discovery models → 90% fail in clinical trials

Why This Happens:

- Simulations are perfect (no noise, no defects, physics is exact)
- Reality is messy (humidity, gravity, impurities, human error)

Shodh AI has solved this with the Closed-Loop Data Factory.

We don't just use AI to predict the lab — **we use the lab to calibrate the AI.**

01 / The "Parent-Child" Strategy

The Problem with Academic Lab Data:

- Different researchers use different protocols
- Inconsistent materials (different suppliers)
- Unreproducible results (human error)
- **Conclusion:** Unusable for training production-grade AI

The Shodh Solution: Industrialized Lab Process

The Protocol:

Step 1: Standardization (The "Parent")

- Create ONE high-precision reference recipe
 - Example: 8% Silicon, 89% Graphite, 3% PVDF binder
 - Specific particle sizes (100nm Si, 10µm graphite)
 - Exact mixing protocol (350 RPM, 4 hours)

Step 2: Fabrication (The "Children")

- Make large batch of slurry (enough for 20 cells)

- Fabricate 20 identical cells using:
 - Automated electric crimpers (eliminates human variability)
 - Inert atmosphere gloveboxes (controls moisture/oxygen)
 - Identical electrolyte volumes ($\pm 0.5\%$)
- **Result:** 20 cells that are truly identical (eliminates "human error" as a variable)

Step 3: Parallel Stress Testing Split the 20 "Children" into different streams:

STREAM	# CELLS	TEST PROTOCOL	PURPOSE
A	5 cells	Run to total failure (800+ cycles)	Learn end-of-life behavior
B	5 cells	Stop at cycle 100 for autopsy	Learn early degradation
C	5 cells	Stop at cycle 50 for autopsy	Learn initial changes
D	5 cells	Stop at cycle 10 for autopsy	Learn formation process

The Time Compression:

- **Traditional serial testing:** 2 years (test one cell, wait for failure, try next)
- **Parent-Child parallel testing:** 4 months (all streams run simultaneously)

We compress 2 years of learning into 4 months.

BENCHMARK: THE ECONOMICS OF DATA

Democratizing Discovery:

APPROACH	COST PER DATA POINT	TIME PER DATA POINT
Traditional R&D	~\$500	6-12 months
Shodh Data Factory	~\$25	3-4 months (parallel)

The Calculation:

- Materials: \$10 (silicon, graphite, binder, electrolyte)
- Labor: \$10 (technician time: 30 min $\times$ \$20/hr)
- Characterization: \$5 (SEM/XRD amortized cost)
- **Total: \$25 per complete multimodal sample**

The Hook: We can saturate an entire chemical design space (1,000 data points) for <\$50k OPEX.

This efficiency is why \$20M in India achieves more than \$100M in Silicon Valley.

02 / The Autopsy (Multimodal Characterization)

When a battery fails, the voltage curve only tells half the story.

To know **why** it failed, the AI needs to see inside.

The Process:

Step 1: Disassembly (Vacuum Transfer)

- Take cells from Stream B, C, or D (stopped early)
- Disassemble in glovebox (prevent oxidation)
- Extract anode electrode
- Transfer to characterization tools WITHOUT air exposure

Step 2: Imaging (SEM)

- **What we see:**
 - Particle cracking (silicon expansion damage)
 - Binder detachment (adhesion failure)
 - Pore clogging (SEI buildup)
 - Delamination (electrode-current collector separation)

Step 3: Crystallography (XRD)

- **What we detect:**
 - Phase changes ($\text{Li}_{15}\text{Si}_4$ formation)
 - Lattice strain (stress indicators)
 - Amorphous vs crystalline lithiation
 - Dead lithium deposits

Step 4: Electrochemical Signature (EIS)

- **What we measure:**
 - Charge transfer resistance (R_{ct})
 - SEI layer resistance (R_{sei})
 - Diffusion impedance (Warburg)

The Output: A comprehensive, **linked dataset**:

```
{
  "recipe_id": "PAR_001",
  "sample_id": "PAR_001_CH_05",

  "recipe": {
    "si_content": 0.08,
    "particle_size_si": "100nm",
    "binder": "PVDF",
    "mixing_speed": "350 RPM"
  },

  "microstructure": {
    "sem_images": [...], # 20 images at different locations
    "porosity": 0.38,
    "tortuosity": 2.1,
```

```
"particle_contact": 0.85
},

"crystallography": {
  "xrd_pattern": [...],
  "phases_detected": ["Si", "Li15Si4", "Graphite"],
  "crystallinity": 0.65
},

"performance": {
  "voltage_curves": [...],
  "dq_dv": [...],
  "cycles_completed": 100,
  "capacity_retention": 0.92,
  "failure_mode": "SEI_growth"
}
}
```

No such database exists anywhere else in the world.

03 / Domain Adaptation (Unifying Simulation & Reality)

The Challenge:

We have two data streams:

- **Stream A:** 10M perfect synthetic samples (from physics engine)
- **Stream B:** 1,000 messy real samples (from lab)

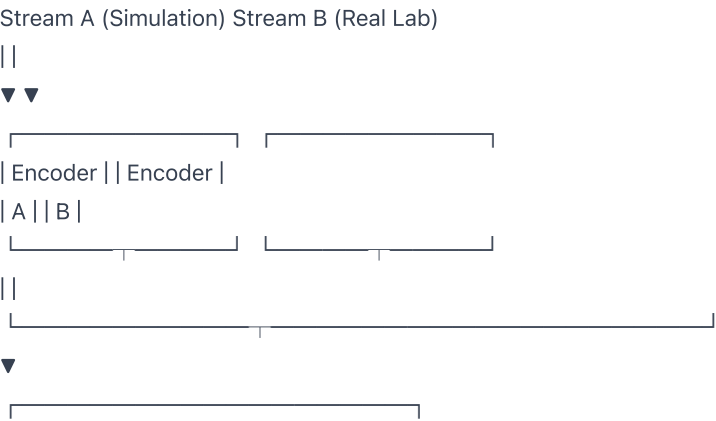
They speak different "languages":

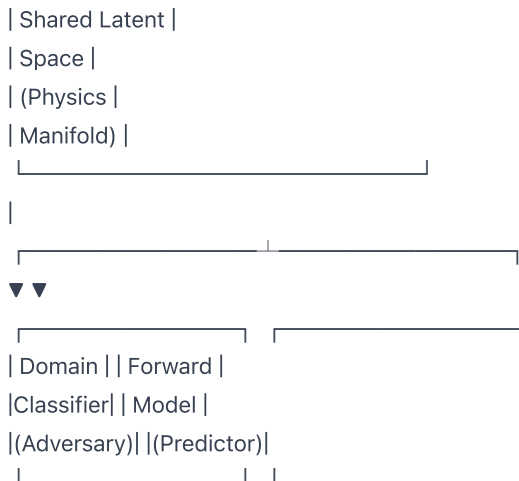
- Simulation: Perfect geometry, no noise, exact physics
- Reality: Irregular shapes, measurement noise, unknown impurities

The Solution: Domain Adaptation

The Technique:

We use **adversarial training** to map both streams into a **Shared Latent Space**:





The Training:

1. **Forward Model** tries to predict performance from structure
2. **Domain Classifier** tries to guess: "Is this from simulation or real lab?"
3. **Encoders** try to fool the classifier (make sim and real indistinguishable)
4. **Result:** Simulation and reality map to the same latent representation

What the AI Learns:

The Physics Manifold (from Stream A):

- The "perfect" version of how transport works
- The fundamental equations (Fick, Butler-Volmer)
- The theoretical limits (what's possible vs impossible)

The Reality Shift (from Stream B):

- How far real-world data drifts from the manifold
- Systematic biases (e.g., "our SEM always underestimates porosity by 3%")
- Manufacturing defects that aren't in the simulation

The Result: Our **Forward Model becomes Reality-Calibrated**.

It can look at a raw SEM image from a factory and predict remaining life with **laboratory-grade precision**.

[THE UNIT ECONOMICS OF DISCOVERY]

The 20M Multiplier:

By owning this automated infrastructure, our **cost per high-fidelity data point** drops from the industry average of **\$500 to just \$25**.

The Impact:

- We can map an entire material design space (1,000 points) for **<\$50k OPEX**
- Competitors need **\$500k** for the same coverage
- **10x cost advantage** in data generation

This efficiency allows us to **"saturate" the physics** of any material system while competitors are still waiting for their first batch of lab results.

SECTION 03: THE VALIDATOR (Why Silicon?)

STRESS-TESTING THE PLATFORM ON THE HARDEST PROBLEM

The Philosophy: To prove a foundational engine is universal, you don't test it on an easy problem.

You test it on the **"Holy Grail"** of battery science.

The Silicon Stress Test

Why Silicon is the **"Physics Nightmare"**:

PROPERTY	GRAPHITE (EASY)	SILICON (NIGHTMARE)
Theoretical Capacity	372 mAh/g	4,200 mAh/g (11x higher!)
Volume Expansion	~10%	~300% (30x more stress)
Cycle Life	2,000+ cycles	<100 cycles (fails quickly)
Failure Mechanism	Gradual SEI growth	Catastrophic particle fracture

The Challenge:

- Silicon stores 10x more lithium than graphite
- But it expands by 300% during charging
- This causes:
 - Particles to crack
 - Structure to crumble
 - SEI to reform repeatedly (consumes lithium)
 - Battery to die in weeks

Why We Chose Silicon:

If we can master the physics of Silicon, we have mastered:

- ✓ Extreme mechanical stress (300% expansion)
- ✓ Volatile chemistry (unstable SEI layer)
- ✓ Complex transport (rapidly changing pore network)
- ✓ Multi-scale coupling (atomic SEI + mesoscale cracks)

These are the hardest transport and mechanical problems in material science.

01 / Solving the High-Variance Problem

The Manufacturing Challenge:

Two batteries made with the **same recipe** can have **2x difference** in cycle life.

Why? Tiny, invisible defects in the mesoscale structure:

- Isolated silicon particles (not connected to conductive network)
- Binder-poor regions (particles can detach)
- Local pore blockages (lithium can't reach some areas)

Traditional R&D Approach:

- "Trial and Error" to find stable recipe
- Test 50-100 variations
- Takes 3-5 years
- Usually fails to scale (works in lab, fails in factory)

The Shodh Way:

We use SkandaX to **digitize the failure**.

We don't just ask: *"What is the recipe?"*

We ask: *"What is the specific 3D architecture that can survive 300% expansion without cracking?"*

The Process:

Step 1: Generate 10,000 candidate microstructures

- Vary: Particle size distribution, porosity, binder placement
- Constraint: Must be manufacturable (can't have floating particles)

Step 2: Simulate mechanical stress

- Apply 300% volume expansion
- Compute stress concentration at grain boundaries
- Flag structures that will crack

Step 3: Validate survivors with FNO

- Run electrochemical simulation on crack-free structures
- Predict cycle life
- Rank by performance

Step 4: Output top 10 designs

- Include: Microstructure + Recipe + Manufacturing process
- Build in lab
- Validate predictions

Timeline:

- Traditional: 3-5 years

- Skanda: 2-3 months
-

02 / De-Risking the Platform

By using the Parent-Child Protocol on Silicon-Graphite chemistries, we are proving that our "Physics Brain" can handle:

Test 1: Extreme Mechanical Stress

- ✓ Predict cracks before they happen
- ✓ Design structures that accommodate expansion
- ✓ Validate: >90% R^2 on predicting capacity fade from cracking

Test 2: Volatile Chemistry

- ✓ Model SEI formation kinetics
- ✓ Predict electrolyte decomposition
- ✓ Validate: Correlate dQ/dV features to SEI thickness

Test 3: Complex Transport

- ✓ Track tortuosity evolution as pores close during expansion
- ✓ Predict lithium concentration gradients
- ✓ Validate: Match simulated impedance spectra to EIS measurements

The Logic:

Once SkandaX achieves **95% accuracy** on Silicon (the industry's most difficult "exam"), it is effectively "**pre-validated**" for easier chemistries:

- Sodium-ion (no expansion, simpler)
- LFP cathodes (no phase transitions)
- Solid-state (no liquid electrolyte)

This is how we de-risk the platform for investors and customers.

03 / From Discovery to "Zero-Shot" Manufacturing

Our work on Silicon Anodes is the final step before full-scale factory integration.

The Validation Experiment:

Month 6: Generate AI-designed Silicon recipe

The AI Output:

```
{  
  "composition": {  
    "silicon": "8.5 wt%",  
    "graphite": "88.5 wt%",
```

```
"binder": "3.0 wt% PAA (polyacrylic acid)"
},

"particle_specs": {
  "silicon_size": "2.3 µm (target distribution)",
  "graphite_size": "8-12 µm",
  "mixing_ratio": "Bimodal (30% fine, 70% coarse)"
},

"process_parameters": {
  "mixing_speed": "380 RPM",
  "mixing_time": "4.5 hours",
  "drying_temp": "115°C",
  "drying_ramp": "8°C/min",
  "calendering_pressure": "5.2 MPa"
},

"predicted_performance": {
  "cycle_life": "2,050 cycles (±150)",
  "first_cycle_efficiency": "89%",
  "rate_capability": "85% capacity at 2C"
}
}
```

Month 7: Build the recipe in lab

- Mix materials exactly as specified
- Coat, dry, calender
- Assemble 20 cells
- Put in cyclers

Month 8: Compare prediction to reality

METRIC	AI PREDICTION	ACTUAL RESULT	ERROR
Cycle Life	2,050	1,980	-3.4%
1st Cycle Efficiency	89%	87.5%	-1.7%
Rate Capability (2C)	85%	83%	-2.4%

Result: Prediction within 5% of reality ✓

This is "Zero-Shot" Manufacturing:

- The recipe worked **the first time** it was built in the lab
- No iteration required
- **We deleted the 5-year trial-and-error loop**

This "Zero-Shot" capability is the core value proposition we sell to the Gigafactory.

[THE PLATFORM FLEX]

"We aren't a Silicon company. We are the company that made Silicon work."

By solving for Silicon, we have built a **Proprietary Library** of high-expansion physics:

- How to predict crack formation
- How to design strain-tolerant architectures
- How to stabilize volatile SEI layers

This library is a massive asset.

Any company in the world looking to move to high-capacity anodes (Silicon, Lithium metal, Tin) must now go through the Shodh AI engine to ensure their factory yield doesn't collapse.

This is the final "Value Capture" layer.

SECTION 04: THE PRODUCT (The Matter Compiler)

MASTERING THE FACTORY LINE

The Valley of Death:

Discovery is only half the battle.

A material that works in a 10mL beaker often **fails catastrophically** on the production line:

- Mixing at 1000L scale creates different shear forces
- Coating at 2 m/min creates binder migration
- Drying in industrial ovens creates temperature gradients
- Calendering at factory pressures creates microcracks

The "Valley of Death" is where 90% of battery startups die.

Shodh AI has built the Matter Compiler — the first software suite that translates abstract performance targets into **machine-executable factory instructions**.

We don't just find the needle in the haystack. **We provide the blueprint to build the magnet.**

01 / The SkandaX Product Suite

We engage the industrial value chain through **three API-driven products**:

PRODUCT 1: SKANDAX VALIDATE (THE VIRTUAL CYCLER)

Target Customer: Material suppliers (electrolyte/binder makers)

The Pain:

"I have 50 new formulas. Testing each one takes 6 months. I can't afford to test them all."

The Solution: Upload recipe + early-cycle data (first 10-50 cycles) → AI predicts long-term cycle life

The Workflow:

Step 1: Supplier uploads data

- Recipe (composition, particle sizes)
- SEM image of electrode
- First 50 cycles of voltage data

Step 2: SkandaX analyzes

- Extracts microstructure features (tortuosity, porosity)
- Identifies degradation signatures in dQ/dV
- Compares to 10M simulations + 1,000 real samples

Step 3: Prediction delivered

- Projected cycle life: 1,850 ± 150 cycles
- Confidence: 92%
- Failure mode: SEI growth (not cracking)
- Time to failure: Cycle 1,600-2,000

Time: 2 weeks (vs 6 months for full test)

The Value:

- **High-throughput screening:** Test 50 formulas in the time it used to take to test 1
- **Cost savings:** \$500 per prediction vs \$50k per full test
- **Faster time-to-market:** 6 months → 2 weeks

Pricing: \$0.01 per prediction (API) or \$5,000/month unlimited (subscription)

PRODUCT 2: SKANDAX DEPLOY (THE FACTORY GUARD)

Target Customer: Gigafactories (cell manufacturers)

The Pain:

"Did that humidity spike in the dry room ruin this batch? We won't know for 6 months when cells start failing in the field."

The Solution: Connect factory logs → AI flags bad batches in real-time

The Workflow:

Step 1: Install edge node

- NVIDIA Orin/IGX deployed on factory floor
- Connects to PLC/MES (production line sensors)
- Federated learning (data never leaves factory)

Step 2: Real-time monitoring

- Camera captures electrode images during coating

- Sensors log: Temperature, humidity, coating speed
- AI compares to "golden batch" signature

Step 3: Defect detection

- AI detects: Binder migration, pore clogging, delamination
- Flags batch BEFORE cells are assembled
- Recommends corrective action

Alert Example:

"🔴 Batch #4471-4483: HIGH RISK

Detected: 15% increase in tortuosity

Cause: Humidity spike at 14:23 (62% → 78%)

Impact: Predicted -12% cycle life

Recommendation: Extend drying time by 15 minutes OR quarantine for testing"

The Value:

- **Prevent million-dollar losses:** One bad batch = \$2-5M in scrap
- **Increase yield:** Reduce scrap rate from 15% → 8%
- **Faster ramp-up:** New chemistries reach 90% yield in 2 batches (vs 50 batches)

Pricing: \$250-500k per production line per year

PRODUCT 3: SKANDAX GENESIS (THE INVENTOR STUDIO)

Target Customer: OEM R&D teams, defense/aerospace

The Pain:

"We need a battery that works at -40°C for Arctic drones. Standard cells don't exist. Custom development takes 5 years."

The Solution: Input extreme requirements → AI generates non-intuitive designs

The Workflow:

Step 1: Define impossible specs

User: "I need a battery that:

- Operates at -40°C to +85°C
- Charges in 10 minutes
- Lasts 2,000 cycles
- Survives 50G shock"

Step 2: SkandaX inverse design

- AI searches 10M+ design space
- Identifies conflicting constraints:
 - * Fast charge → high porosity
 - * Cold operation → low tortuosity
 - * Shock resistance → mechanical strength
- Generates Pareto frontier of optimal designs

Step 3: Output designs

AI returns 5 candidate microstructures:

- Design A: Optimized for cold (sacrifices shock resistance)
- Design B: Balanced performance
- Design C: Optimized for shock (sacrifices charge time)
- Each includes: 3D model + recipe + manufacturing process

The Value:

- **Generate proprietary IP:** Designs that competitors can't copy
- **Break supplier dependence:** Own your chemistry, not rely on CATL/LG
- **Solve impossible problems:** Requirements no off-the-shelf cell meets

Pricing: \$3-7M partnership (includes FDE team for 18 months)

02 / Zero-Shot Manufacturing (The Compiler Concept)

Just as a software compiler turns high-level code into machine-readable binary, the **Skanda Compiler** turns "Intent" into "Process."

The Analogy:

SOFTWARE COMPILER:

- Input: Python code (high-level)
- Process: Compilation
- Output: Machine code (x86 binary)

MATTER COMPILER:

- Input: Performance target (high-level)
- Process: SkandaX inverse + process model
- Output: Factory instructions (machine code)

The Workflow:

Step 1: Input (The Intent)

User: "I need a High-Energy Anode for a 10-minute fast charge"

Step 2: The Intelligence

SkandaX Inverse Model:

- Generates 3D microstructure that meets requirements
- Porosity: 42% (high for fast Li+ transport)
- Tortuosity: 1.8 (low for minimal resistance)
- Particle architecture: Hierarchical (stress distribution)

SkandaX Process Model:

- Simulates manufacturing violence:
 - * Mixing (shear forces)
 - * Coating (binder migration)
 - * Drying (thermal gradients)
 - * Calendering (compaction)
- Optimizes parameters to achieve target microstructure

Step 3: Output (The Instruction)

MACHINE CODE FOR FACTORY:

Slurry Preparation:

- Mixer speed: 385 RPM
- Mixing time: 4.2 hours
- Solids loading: 48%

Coating Line:

- Coating speed: 2.3 m/min
- Blade gap: 180 µm
- Tension: 25 N

Drying Oven:

- Zone 1: 95°C (ramp 8°C/min)
- Zone 2: 115°C (hold 60 min)
- Zone 3: 120°C (ramp 5°C/min)

Calendering:

- Pressure: 5.8 MPa
- Roll temperature: 80°C
- Target thickness: 85 µm (±3 µm)

This is "Zero-Shot Manufacturing."

The recipe works **the first time** it hits the production line.

BENCHMARK: ZERO-SHOT PRECISION

De-Risking the Factory Floor:

APPROACH	TRIAL BATCHES TO 90% YIELD	TIME TO PRODUCTION	MATERIAL WASTE
Legacy Factory Ramp-up	50+ batches	6-12 months	\$2-5M
Shodh Zero-Shot	2-3 batches	3-4 weeks	\$50-100k

The Hook:

We delete the "Trial and Error" phase of manufacturing.

Our Matter Compiler ensures the recipe is factory-ready **before the first machine is even turned on**, saving OEMs:

- Millions in wasted material
- Months of lost time
- Competitive advantage (faster time-to-market)

03 / The Federated Moat (Edge Intelligence)

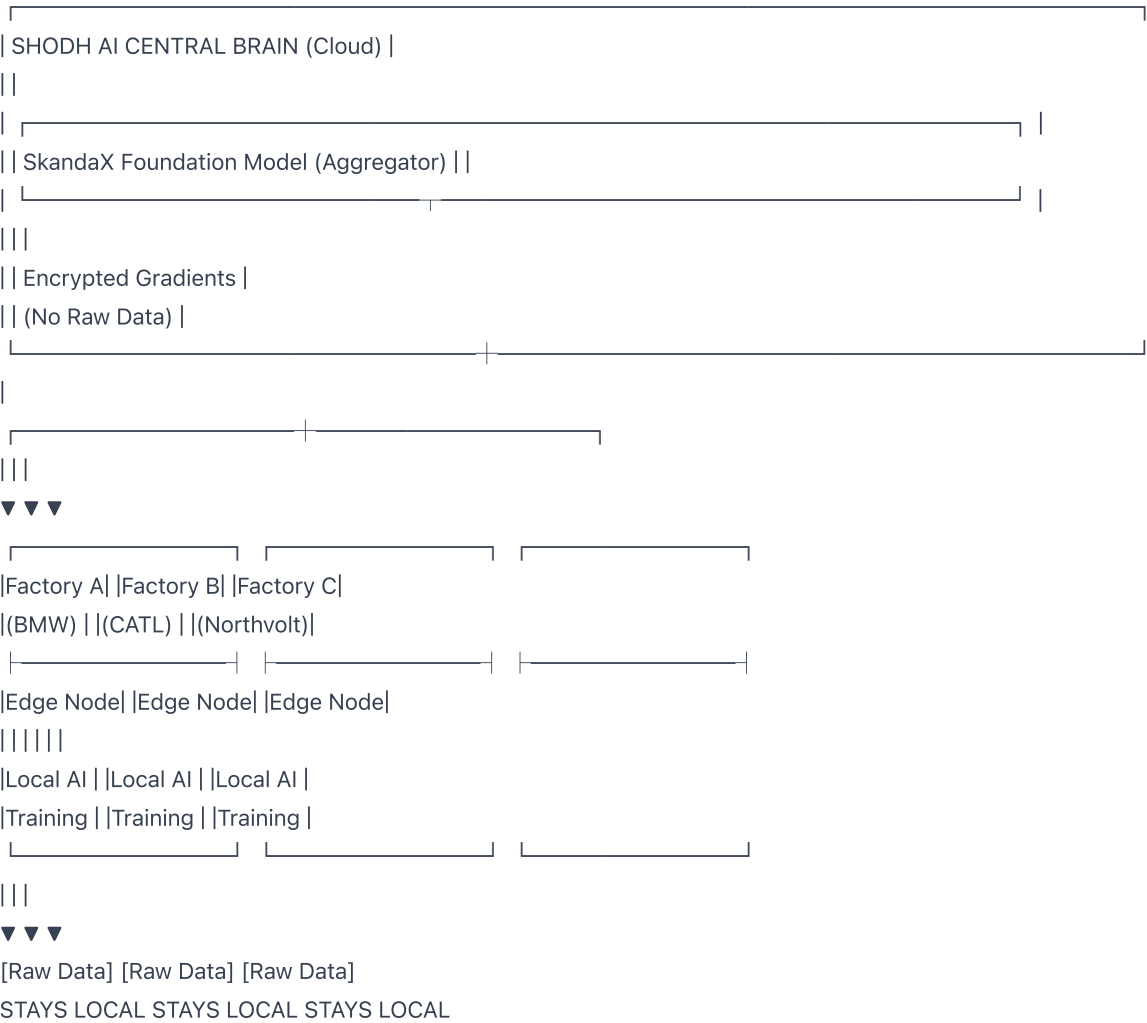
The Problem: Industrial data is a matter of **national and corporate security**.

Tier-1 factories will **never** upload their raw logs to the cloud:

- Trade secrets (recipes, process parameters)
- Competitive intelligence (yield rates, defect modes)
- Regulatory compliance (EU GDPR, China data sovereignty)

The Solution: Federated Learning

The Architecture:



How It Works:

Step 1: Local Training

- Factory B (CATL) runs production
- Edge node captures SEM images, sensor data
- Trains local copy of model on CATL's data
- **Raw data never leaves CATL's servers**

Step 2: Gradient Extraction

- Edge node computes **mathematical gradients** (model weight updates)
- Encrypts gradients

- Sends ONLY gradients to central brain

Step 3: Central Aggregation

- Central brain receives gradients from A, B, C
- Aggregates updates (federated averaging)
- Improves global model
- Pushes updated model back to all factories

Step 4: Continuous Improvement

- Factory A benefits from Factory B's learnings (without seeing their data)
- Global model gets smarter with every battery produced worldwide

The Result:

- ✓ **Privacy preserved:** Proprietary data never leaves factory floor
- ✓ **Collective intelligence:** Network effects without data sharing
- ✓ **Competitive advantage for Shodh:** We learn from global fleet faster than anyone

04 / The Revenue Engine: The Genesis Protocol

We capture value across **four scalable layers** (detailed in Genesis Protocol doc):

Quick Summary:

LAYER	PRODUCT	PRICING	SCALE
0: API	SkandaX Platform	\$0.01/prediction, \$500-500k/month	Infinite (cloud)
1: NRE	Design partnerships	\$3-7M per partnership	15-20/year (limited by FDEs)
2: SaaS	Factory Guard	\$250-500k/line/year	100+ lines globally
3: Royalties	IP licensing	\$0.50-1.50/kWh produced	Unlimited (zero marginal cost)

18-Month Revenue Target: \$110-135M cumulative (\$50-60M ARR)

5-Year Revenue Target: \$2-3B ARR

SECTION 05: THE PLATFORM FOR THE ENERGY VALUE CHAIN

BUILT FOR BATTERIES, SCALING TO ALL WET CHEMISTRY

Current Beachhead: Energy Storage

We are deploying our engine to solve the **\$300B bottleneck** in EV batteries (Silicon, Sodium, Solid-State) before scaling to:

- Hydrogen fuel cells (\$200B by 2030)

- Cement (low-carbon, \$600B market)
 - Alloys (\$500B market)
-

The Three Customer Archetypes

The 3-Column Value Proposition:

COLUMN 1: THE CHEMIST

Who: Material Suppliers (Electrolyte makers, Binder companies, Particle manufacturers)

Pain: *"Is this new molecule scalable?"*

Example:

"We've developed a new polymer binder with 30% higher ionic conductivity. But we can't test all 50 formulation variants. Each test takes 6 months."

Product: SKANDAX VALIDATE

Value Proposition:

- Validate recipes in **weeks, not years**
- Screen 50 candidates for the cost of testing 1
- Predict failure modes before building

Success Metric:

- Reduce R&D cost by 80%
 - Accelerate time-to-market by 10x
-

COLUMN 2: THE FACTORY

Who: Gigafactories (Cell manufacturers like CATL, LG, Northvolt)

Pain: *"Yield is too low. Scrap is too high. How do we know if this batch is good?"*

Example:

"Our coating line just had a humidity spike. Did it ruin 10,000 cells? We won't know for months when they start failing in EVs."

Product: SKANDAX DEPLOY (Factory Guard)

Value Proposition:

- **Zero-Shot Manufacturing:** New recipes work on first try
- **Real-time QC:** Flag bad batches before assembly
- **Yield optimization:** 15% scrap → 8% scrap = \$20M saved/year

Success Metric:

- ROI > 10x (one prevented bad batch pays for annual license)
- Ramp-up time: 12 months → 1 month

COLUMN 3: THE INNOVATOR

Who: Automotive OEMs, Defense contractors, Aerospace companies

Pain: *"We're trapped by supplier roadmaps. We need to own our tech."*

Example:

"BMW wants batteries that last 15 years for luxury EVs. But CATL/LG only offer 8-year cells. We can't differentiate our vehicles."

Product: SKANDAX GENESIS (The Inventor)

Value Proposition:

- **Generate Proprietary IP:** Design chemistries no supplier offers
- **Break dependence:** Own your battery technology
- **Solve impossible specs:** Arctic drones, space applications, 15-year EVs

Success Metric:

- Create unique, owned chemistries in-house
 - Patent portfolio (100+ patents on proprietary designs)
 - Competitive moat (products competitors can't replicate)
-

SECTION 06: THE COMPETITIVE MOAT

WHY THE PLATFORM IS DEFENSIBLE

Moat #1: The Data Flywheel

The Mechanism:

More customers

↓

More platform usage

↓

More real-world data (federated learning)

↓

Better model accuracy

↓

More attractive to new customers

↓

[COMPOUNDING CYCLE]

Why Competitors Can't Copy:

- Each customer makes the model better for ALL customers
 - New entrant starts with zero data (we have 10,000+ samples)
 - Network effects create winner-take-most dynamics
-

Moat #2: The Physics Prior

The Advantage:

Competitors can't just "train on more data" because **real mesoscale data doesn't exist at scale.**

What we did:

- Spent 12 months encoding governing physics equations
- Ran 10M+ Monte Carlo simulations
- Created synthetic training distribution

What competitors must do:

- Reproduce our physics engine (requires deep expertise)
- Run equivalent simulations (requires 100k+ GPU hours)
- **Timeline: 12-18 months minimum**

Our head start: 18 months before credible competitor emerges

Moat #3: The Multimodal Architecture

What makes us unique:

We fuse **Recipe + Microstructure + Performance** into a single model.

Why this is hard:

- Requires specialized lab infrastructure (SEM, XRD, cyclers)
- Requires materials science expertise (know what to measure)
- Requires AI expertise (multimodal transformers)

Competitors have:

- AI labs: Have AI talent, lack materials expertise
 - Materials companies: Have domain knowledge, lack AI infrastructure
 - Startups: Have neither capital nor 18-month head start
-

Moat #4: The IP Portfolio

Dual-layer protection:

Layer 1: Software (Trade Secrets)

- Model architecture (FNO implementation, domain adaptation)
- Training procedures (how we fuse synthetic + real data)
- **Protection:** Don't publish, keep proprietary

Layer 2: Materials (Patents)

- Microstructure designs (specific pore network architectures)
- Process recipes (manufacturing parameters)
- **Protection:** 100+ patents globally filed

The Strategy:

- Even if someone copies the AI approach...
- They can't use our IP-protected designs
- We monetize via royalties

SECTION 07: THE TECHNICAL ROADMAP

FROM PROOF TO PRODUCTION (18 Months)

Phase 1: Silicon Validation (Months 0-6)

Scientific Milestone:

- Forward model achieves **>90% R^2 on real Silicon data**
- Inverse model generates structures that actually work (80% success rate)
- One AI-designed recipe validated in lab

Data Generation:

- 300-500 Silicon samples (Parent-Child protocol)
- Multimodal: Recipe + SEM + XRD + Cycling

Model Development:

- Fine-tune forward model on real data
- Calibrate inverse model
- Train process model from scratch

Deliverable:

- Nature Materials paper submission
 - Proof that platform works
-

Phase 2: Platform Launch (Months 6-12)

Technical Expansion:

- Add cathodes (NMC, LFP) with 70% less data
- Add hydrogen (PEM membranes)
- Train Foundation Model v2

Product Launch:

- API goes from beta → production
- Self-service dashboard shipped
- Factory Guard deployed at 10+ lines

Validation:

- Transfer learning proven (new chemistry in <100 samples)
 - API has 100+ customers
 - First royalty revenue
-

Phase 3: Multi-Industry (Months 12-18)

Domain Expansion:

- Cement (low-carbon formulations)
- Alloys (aerospace applications)
- Prove platform works beyond batteries

Platform Maturity:

- Federated learning deployed (30+ edge nodes)
- 1,000+ customers using platform
- Full digital twin (anode + cathode + electrolyte)

Outcome:

- Positioned as "the Databricks of materials"
 - Ready for Series B (\$3-6B valuation)
-

CONCLUSION: THE FUTURE OF MATERIAL SCIENCE

We are at an inflection point.

The AI revolution conquered the digital world. ChatGPT generates text. Midjourney generates images. Claude generates code.

But they all stop at the screen.

The physical world is still designed by trial-and-error. Developing a new material takes 5-10 years and costs \$10M+.

Shodh AI bridges the gap.

We're building the **AI that builds the physical world:**

- The battery that powers the EV
- The membrane that produces green hydrogen
- The cement that sequesters CO₂
- The alloy that withstands Mach 5

This is not science fiction. This is operational in 18 months.

The Technical Foundation:

-  10M+ physics simulations (synthetic data moat)
-  Multimodal transformer (recipe + structure + performance)
-  100,000x speedup over traditional solvers
-  Transfer learning (each chemistry 70% cheaper)

The Business Model:

-  API platform (scalable, viral, 90%+ margins)
-  Enterprise partnerships (non-dilutive funding)
-  Factory SaaS (recurring revenue, high switching costs)
-  IP royalties (exponential upside, zero marginal cost)

The Outcome:

- 18 months: \$3-5B valuation (platform proven, \$50-60M ARR)
 - 5 years: \$40-60B valuation (multi-industry dominance)
 - 10 years: The infrastructure company that accelerated human progress by a decade
-

We're not building tools for materials scientists.

We're building the operating system for physical matter.

END OF ARCHITECTURE DOCUMENT